

# Phase 8 - Project Demonstration

PocketSmart AI: Your Smart Budget & Recommendation Assistant

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## 1. Demonstration Overview

The demonstration presents PocketSmart AI from user entry through recommendation generation. It should show the interface, authentication, three planners, AI output and recommendation history.

## 2. Demonstration Objectives

- Show the main interface.
- Demonstrate registration and login.
- Demonstrate Home Planner.
- Demonstrate Party Planner.
- Demonstrate Jewelry Planner with optional image.
- Show recommendation output.
- Show dashboard and history.
- Explain frontend, FastAPI, Gemini and database flow.

## 3. Environment

Run the application using the configured Python/FastAPI environment with required dependencies and environment variables.

# **Project Demonstration - Authentication and Home**

## **4. Step 1 - Register**

Open Register, enter required credentials and submit the form to create the user account.

## **5. Step 2 - Login**

Open Login and enter valid credentials. Demonstrate successful authentication and access to protected dashboard/planner functionality.

## **6. Step 3 - Dashboard**

Show recent recommendations, saved queries and personalized information as implemented by the project.

## **7. Step 4 - Home Planner**

- Open Home Budget Planner.
- Enter total budget.
- Select room type such as Living Room, Kitchen or Bedroom.
- Enter quantities for required items.
- Submit request.
- Show AI-generated home recommendations.

# **Project Demonstration - Party and Jewelry**

## **8. Step 5 - Party Planner**

- Enter total budget and guest count.
- Select event type.
- Provide venue details.
- Submit request.
- Show catering, venue, decoration, entertainment and related suggestions.

## **9. Step 6 - Jewelry Planner**

- Enter budget, occasion and style preference.
- Optionally upload an outfit image.
- Submit request.
- Show recommendations matched to occasion, outfit, style and budget.

## **10. AI Explanation**

Gemini 1.5 Flash Pro processes text, numerical context and optional image input to generate recommendation content. The backend returns the structured result to the frontend.

# **Project Demonstration - History and Technical Flow**

## **11. Step 7 - Recommendation History**

Open Recommendation History and demonstrate the log of previous recommendation queries and results for review or reuse.

## **12. Technical Flow**

- Frontend collects user data.
- FastAPI receives and routes the request.
- Authentication/session identifies the user where required.
- Schema/application logic processes input.
- Recommendation service prepares AI context.
- Gemini processes budget and context.
- Catalog/platform information is used where available.
- Frontend displays recommendations.
- History stores relevant previous queries/results.

## **13. Fallback Demonstration**

If the AI service is unavailable or returns insufficient information, demonstrate the fallback/default recommendation behaviour specified by the project.

## **14. Conclusion**

The demonstration shows PocketSmart AI as an end-to-end intelligent budgeting assistant across Home Interior, Party Planning and Jewelry selection.